AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM254
Managing IT: the why, the what and the how

Final Examination
Fall Semester 2025/2026

Date: 12/Jan/2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: MCQ [20 Marks]

Each question carries a weight of 2 marks

Question 1: [20 marks] Read the following questions and choose the best answer.

1- Which of the following best describes an insertion anomaly in a database?

A. Deleting a record accidentally removes other unrelated data
B. Updating one attribute requires updating multiple records
C. Certain information cannot be added without the presence of other data
D. Data is duplicated across multiple tables

2- What is the intersection of Set A = {a, b, c, d} and Set B = {a, c, e, g, h}?

A. {b, d, e}
B. {a, b, c, d, e, g, h}
C. {a, c}
D. {e, g, h}

3- Which of the following is NOT a rule for implementing controlled sharing in a database?

A. There is no sharing
B. Give users more privileges than they need
C. Limit access to the DBMS
D. Limit access to individual data objects

4- Which software development approach is most suitable when requirements are not fully clear at the start?

A. Waterfall
B. Spiral
C. Iterative
D. Incremental

5- Users, sponsors, and other members of the organization are known as:

A. Team stakeholders
B. Organizational stakeholders
C. Contracted stakeholders
D. External stakeholders

6- Which type of team is cross-functional, with each member participating in all aspects of software development, including coding?

A. Agile team
B. Incremental team
C. Iterative team
D. Waterfall team

7- Which SQL language is used to create tables and define the physical schema of a database?

A. Data Manipulation Language (DML)
B. Data Definition Language (DDL)
C. Data Control Language (DCL)
D. Transaction Control Language (TCL)

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8- The work schedule in project management is created using:

A. Activity bar chart
B. Project chart
C. Resource table
D. Activity bar chart and resource table

9- Which type of cost schedule shows the total costs accumulated over time during a project?

A. Cumulative project costs
B. Resource allocation costs
C. Risk-adjusted costs
D. Daily project costs

10- When a particular problem arises during plan execution, which of the following is a valid approach to solve the problem?

A. Work harder
B. Replan
C. Increase resources
D. All of the above

PART 2: Short questions [35 Marks]

Question 2: [6 marks]

a- Explain what risk management is in project management, and discuss why identifying and managing risks early is important for project success.
b- Describe the difference between business risks and project risks.

Question 3: [4 marks]

Explain what project management is and list four key factors that contribute to the success of a project.

Question 4: [3 marks]

Explain when project portfolio management is needed, and describe the difference between a project portfolio and a program.?

Question 5: [7 marks]

The following relation stores information about patients and the medical tests they have taken. The primary keys are **PatientID** and **TestID**.

PatientTests (**PatientID**, **TestID**, TestName, TestDate)

And the instance values:
PatientTests (P101, (301, 302), (Blood Test, X-Ray), (2025-10-01, 2025-10-02))
PatientTests (P102, (301, 304), (Blood Test, CT Scan), (2025-10-03, 2025-10-04))

a) Explain why this relation does not satisfy the second normal form (2NF). (2 marks)
b) Convert this relation to the second normal form (2NF). (5 marks)

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Question 6: [15 marks]

Consider a hospital system database. You are asked to create and manage a table that stores information about doctors.

a) Create a table named Doctors with the following structure: (3 marks)
   - DoctorID (integer, primary key)
   - DoctorName (varchar(50), not null)
   - Specialization (varchar(30))
   - ExperienceYears (integer, must be greater than or equal to 1)
   - Salary (double)

b) Insert the following data into the Doctors table: (3 marks)
   - (1, 'Dr. Ali', 'Cardiology', 5, 2500.00)
   - (2, 'Dr. Hiba', 'Neurology', 8, 3100.00)
   - (3, 'Dr. Karim, 'Pediatrics', 2, 1800.00)
   - (4, 'Dr. Rana', 'Cardiology', 10, 4000.00)

c) Write an SQL statement to display all doctors who have more than 5 years of experience. (2 marks)

d) Update the salary of Dr. Walid to 2000.00. (2 marks)

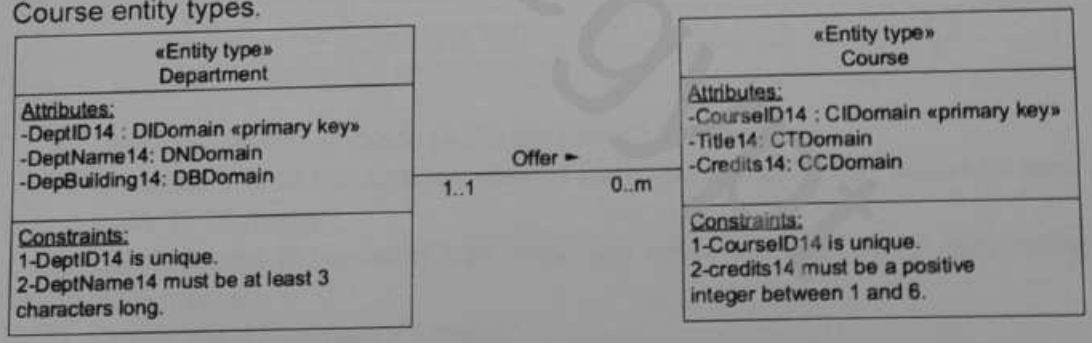
e) Write an SQL query to display the number of doctors in each specialization. (3 marks)

f) Create a new SQL user named user1 with login and password 'Default'. (2 marks)

Part 3: Problem Solving Questions [45 Marks]

Question 7: [18 marks]

The below E-R diagram shows the relationship **Offer** between the Department and Course entity types.



a) Write the textual notation for each entity type, including primary keys, attributes, and constraints. (3 Marks)

b) What is the optionality and multiplicity of **Department** and **Course** in the **Offer** relationship? (3 Marks)

c) Draw the relation-for-relationship representation from the original E-R diagram. (7 Marks)

d) Draw the posted-key representation from the original E-R diagrams. (5 Marks)

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Question 8: [12 marks]

Your task is to create the Entity-Relationship Diagram (ERD) using the below information.

Conceptual Model
Entities: Doctor, Patient, Test.

Relationships:

- Each doctor may perform one or more tests.
- Each test must be performed by one doctor.
- Each patient can take one or more tests.
- Each test must be taken by one patient.

a) Create an Entity-Relationship Diagram (ERD) that represents the relationships described above. (6 marks)
b) For each entity, include two attributes of your choice. (3 marks)
c) Add two constraints per entity to the ERD diagram. (3 marks)

Question 9: [15 marks]

The Marketing Team is planning a new product launch. The project involves several key activities with specified durations and dependencies.

Activity	Task Description	Duration (days)	Predecessor(s)
A	Conduct market research	5	—
B	Develop marketing strategy	4	A
C	Design promotional materials	3	B
D	Prepare digital campaign content	4	B
E	Coordinate with production team	2	A
F	Launch advertising campaign	3	C, D, E
G	Monitor campaign performance	2	E, F

a) Draw the Activity-on-Node (AoN) diagram (3.5 marks)
b) Calculate and fill the duration, earliest start, and earliest finish values for each node. (5.25 marks)
c) Calculate and fill the latest start, float, and latest finish values for each node. (5.25 marks)
d) Identify the critical path. (1 mark)

End of questions

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